Lecture 09
Probabilities and Computing

Algorithms, AI & Data Science II

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Notes:

- **Lecture L09: Probabilities and Computing** 28.05.2025
- **Educational objective:** We study foundations of probabilistic algorithms in data science and machine learning. We cover basic concepts of probability theory, introduce important distributions and explore limit theorems that are used in data science.
  - Foundations of Probability Theory
  - Multivariate distributions
- **Handout version:** *2025/05/26 09:22:53*

Motivation

- we have used formal logics to represent **propositions that are either true of false**
- each assignment of truth values to propositional variables (or *structure* in first-order logic) represents a **different world**
- logical inference allows to check whether statements are **true in every possible world**
- in real-world applications, we must be able to cope with **uncertainty**
- we need **calculus for uncertainty**, i.e. numerical quantification of uncertainty that we call **probability**
- basis for **statistical inference and probabilistic algorithms**

Ying Yang

image credit: Wikimedia commons, public domain

Notes:

Dealing with uncertainty

- in AI and data science, we **use observations to draw conclusions or make predictions**
- we often have **partial knowledge** about the world

examples for uncertainty

1. image data with limited resolution
2. survey of political opinions in Germany
3. outcome of a future dice roll
4. future action of opponent in a game
5. position/momentum of particles

- uncertainty due to **limited precision** measurements, **incomplete data**, **limited predictability of future events**, or **fundamental non-determinism**
- how can we formalize our **lack of knowledge**?

Werner Heisenberg
1901 – 1976

image credit: Bundesarchiv, Bild 183-R57262

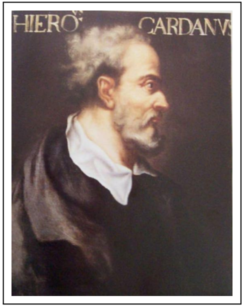
Notes:

Quantifying uncertainty

- we use **probability** to quantify our uncertainty about **events**
- larger probability = event is more likely to happen
- concept of **probability originally based on odds**, i.e. ratio between number of events with a given outcome vs. other outcomes → G Cardano: Liber de ludo aleae (Book on Games of Chance), 1564

> "Knew that we ventured on such dangerous seas
> That if we wrought out life 'twas ten to one"
>
> → William Shakespeare, Henry IV, written before 1598

- odds of $10 : 1$ means that a given **outcome is ten times more likely than other outcomes**



Gerolamo Cardano (1501 – 1574)

image credit: Wikimedia Commons, public domain

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Notes:

Sample space, events, and probabilities

- we define a **universe** Ω of possible, mutually exclusive outcomes, which we call **sample space**
- sample space can be **finite, infinite, discrete, continuous**
- each outcome $\omega \in \Omega$ is a "possible version of reality", i.e. outcomes are **mutually exclusively and exhaustive**
- to each outcome ω, we assign probability $P(\omega) \in [0, 1]$ that captures our uncertainty
  - $P(\omega) = 1 \implies \omega_i$ is **certain**
  - $P(\omega) = 0 \implies \omega_i$ is **impossible**
- any set $E \subseteq \Omega$ of outcomes is called **event**
- we define **additive probability measure on events**, i.e.

$$P(E) = \sum_{\omega \in E} P(\omega)$$

example

- For roll of a dice, sample space Ω contains exactly six **mutually exclusive outcomes** $\omega_i \in \Omega$ defined by possible dice rolls, i.e.

$$\Omega = \{1, 2, \dots, 6\}$$

- For fair dice, we have

$$P(\omega_i) = \frac{1}{6} = \frac{1}{|\Omega|}$$

- $E_1 := \{2, 4, 6\} \subseteq \Omega$ corresponds to **event that dice shows even number**.
- $E_2 := \{1, 2, 3\} \subseteq \Omega$ corresponds to **event that dice shows number smaller than four**.
- $P(E_1) = P(E_2) = 3 \cdot \frac{1}{6} = \frac{1}{2}$

Notes:

Probability spaces

definition

For sample space Ω, event space $\mathcal{F} \subseteq 2^\Omega$ and probability function $P : \mathcal{F} \to [0,1]$ we call triplet $(\Omega, \mathcal{F}, P)$ **probability space** iff

- $\Omega \in \mathcal{F}$
- $E \in \mathcal{F} \implies E^C = \Omega - E \in \mathcal{F}$ → closure under complement
- $E_1, E_2, \ldots \in \mathcal{F} \implies \bigcup_i E_i \in \mathcal{F}$ → closure under union
- For pairwise disjoint events $E_1, E_2, \ldots \in \mathcal{F}$ with $E_i \cap E_j = \emptyset$ we have

$$P\left(\bigcup_i E_i\right) = \sum_i P(E_i)$$

→ countable additivity

- $P(\Omega) = 1$

- $P : \mathcal{F} \to [0,1]$ is **probability measure** such that the measure on whole sample space is one
- from the definition it follows that $P(\Omega^C) = P(\emptyset) = 0$

example

- For roll of a dice, **sample space** Ω contains exactly six mutually exclusive outcomes $\omega_i \in \Omega$ defined by possible dice rolls, i.e.

$$\Omega = \{1, 2, \ldots, 6\}$$

- **event space** $\mathcal{F} = 2^\Omega$ is closed under union and complement
- $E_1 = \{2, 4, 6\} \in 2^\Omega$ is event that dice shows even number
- For fair dice, we have

$$P(\omega_i) = \frac{1}{6} = \frac{1}{|\Omega|}$$

- $P(\Omega) = 6 \cdot \frac{1}{6} = 1$
- $(\Omega, 2^\Omega, P)$ is a probability space

Notes:

Set theory and basic rules of probability

- for **union of events** $A, B \subseteq \Omega$ we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

  where $A \cap B$ is called **joint event**

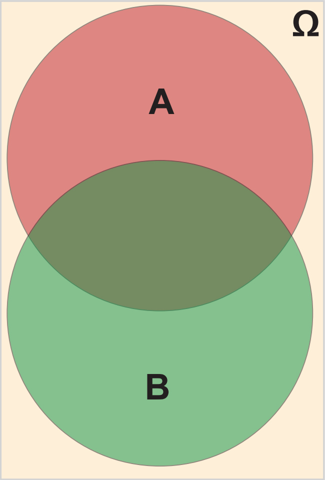
- two events A and B are **mutually exclusive** iff they cannot occur jointly, i.e. iff $A \cap B = \emptyset$

- for mutually exclusive events A, B we have $P(A \cap B) = 0$ and thus

$$P(A \cup B) = P(A) + P(B)$$

- events A and B are **independent** iff $P(A \cap B) = P(A)P(B)$

- different from mutually exclusive events, **independent events can occur at the same time (but do not affect each other)**



Notes:

Prior and posterior probabilities

- probability of event B given **prior knowledge** that A has occurred?
- corresponds to probability of B in **sample space restricted to** A
- **conditional probability** $P(B|A)$ of event B given event A is

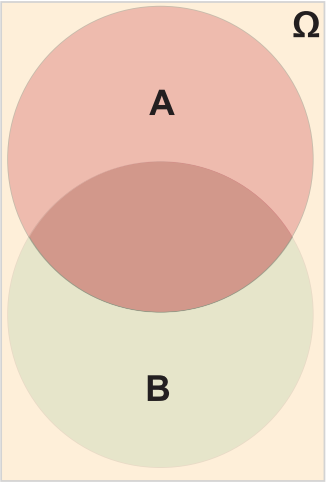
$$P(B|A) := \frac{P(A \cap B)}{P(A)}$$

- we call $P(A)$ **prior** and $P(B|A)$ **posterior probability**
- **chain rule for joint event** $A \cap B$ allows to swap prior, i.e.

$$P(A \cap B) = P(B|A)P(A) = P(A|B)P(B)$$

- for independent events A and B we have

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A) \cdot P(B)}{P(A)} = P(B)$$



Notes:

Bayes rule

Bayes rule

Using the basic rules of probability one can show

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

- Bayes rule captures how prior knowledge about event A **changes our belief** about event B
- basis for **Bayesian inference or learning** $\rightarrow$ L10 Statistical Inference
- highlights Bayesian interpretation of $P(A)$ as our **degree of belief in event** A
- for **independent events** A **and** B we recover

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)} = \frac{P(A \cap B)P(B)}{P(B)P(A)} = P(B)$$

Thomas Bayes
1702 - 1761

image credit: public domain (image might not be authentic)

Notes:

Group Exercise 09-01

1. Consider a random experiment in which we roll a fair dice twice. Define the sample space Ω and formally define event $B \in 2^{\Omega}$ that the sum of both dice rolls is ten. Calculate $P(B)$?

   Sample space Ω is the set of possible outcomes of two dice rolls, which we denote as $\Omega = \{(1,1), \ldots, (6,6)\}$. We have $B = \{(i,j) \in \Omega | i + j = 10\} = \{(4,6), (5,5), (6,4)\}$.

   $$P(B) = \sum_{(i,j) \in B} P(B) = |B| \cdot \frac{1}{36} = \frac{1}{12}$$

2. For the random experiment above, what is the probability that the sum of the two dice rolls is 10, given that we know that the first dice roll is a six?

   With $A = \{(6, i)| \text{ for } i = 1, \ldots, 6\}$ we have the prior probability $P(A) = \frac{6}{36} = \frac{1}{6}$ We further have $A \cap B = \{(6,4)\}$ and thus $P(A \cap B) = \frac{1}{36}$

   $$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{1}{36}}{\frac{1}{6}} = \frac{6}{36} = \frac{1}{6}$$

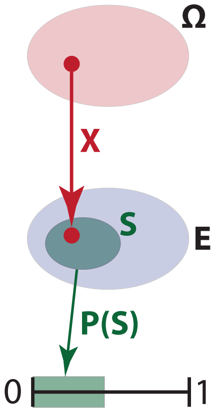
Notes:

Random variables

- for probability space $(\Omega, \mathcal{F}, P)$ and measurable space E we call function $X : \Omega \to E$ **random variable**
- random variable = **function that helps us to quantify random phenomena**
- measure space E can be **discrete or continuous** (e.g. $\mathbb{N}$ or $\mathbb{R}$)
- for any subset $S \subseteq E$ of the measurable space we define

$$P(X \in S) := P(\{\omega \in \Omega | X(\omega) \in S\})$$

- we say X is **random variable with distribution** P and write $X \sim P$



Example

- X is random variable that maps faces of fair dice to measurable space $E := \{1, 2, 3, 4, 5, 6\}$.
- $P(X \leq 3) = \frac{1}{2}$

Notes:

Propositional vs. random variables

- in propositional logic, we can compute truth value of proposition for **specific variable assignment**, e.g.

$$X_0 = X_1 = X_2 = 1$$

- we can use logical inference to make conclusions that hold for **any variable assignment**

- given **partial knowledge** on propositional variables we can assess **probability of truth values of propositions**

exemplary question

- Given $P(X_i = 1) = \frac{1}{2}$ what is the probability that $(X_0 \wedge X_1) \vee X_2$ is True?

- we can map $\wedge$ and $\vee$ to joint event and union of events

X_2	X_1	X_0	$(X_0 \wedge X_1) \vee X_2$
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1
?	?	?	?

Notes:

Group Exercise 09-02

1. For propositional variables X_0, X_1, X_2 consider the truth table of formula $(X_0 \wedge X_1) \vee X_2$. Calculate the probability for possible values of random variable $Y := (X_0 \wedge X_1) \vee X_2$ for $P(X_0 = 1) = P(X_1 = 1) = P(X_2 = 1) = \frac{1}{2}$.

We consider the sample space $\{0, 1\}^3$ and use p_i to denote $P(X_i = 1)$

X_2	X_1	X_0	$(X_0 \wedge X_1) \vee X_2$	$P(X_2 = 1)\cdot$	$P(X_1 = 1)\cdot$	$P(X_0 = 1)$
0	0	0	0	$(1 - p_2)\cdot$	$(1 - p_1)\cdot$	$(1 - p_0)$
0	0	1	0	$(1 - p_2)\cdot$	$(1 - p_1)\cdot$	p_0
0	1	0	0	$(1 - p_2)\cdot$	$p_1\cdot$	$(1 - p_0)$
0	1	1	1	$(1 - p_2)\cdot$	$p_1\cdot$	p_0
1	0	0	1	$p_2\cdot$	$(1 - p_1)\cdot$	$(1 - p_0)$
1	0	1	1	$p_2\cdot$	$(1 - p_1)\cdot$	p_0
1	1	0	1	$p_2\cdot$	$p_1\cdot$	$(1 - p_0)$
1	1	1	1	$p_2\cdot$	$p_1\cdot$	p_0

With $p_i = \frac{1}{2}$ all variable assignments (X_2, X_1, X_0) have probability $\frac{1}{2}^3 = \frac{1}{8}$ and are thus equiprobable. We thus have

$$P(Y = 0) = \frac{3}{8} \text{ and } P(Y = 1) = \frac{5}{8}$$

Notes:

Group Exercise 09-02

- Calculate $P(Y=1)$ and $P(Y=0)$ for $P(X_0=1)=0.9$, $P(X_1=1)=0.2$ and $P(X_2=1)=0.7$.

Mapping $A \vee B$ to $A \cup B$ and $A \wedge B$ to $A \cap B$ and using $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ we have

$$
\begin{aligned}
P((X_0 \wedge X_1) \vee X_2 = 1) &= P(X_0=1) \cdot P(X_1=1) + P(X_2=1) - P(X_0=1) \cdot P(X_1=1) \cdot P(X_2=1) \\
&= 0.9 \cdot 0.2 + 0.7 - 0.9 \cdot 0.2 \cdot 0.7 = 0.754 \\
P(\neg((X_0 \wedge X_1) \vee X_2) = 1) &= P((\neg X_0 \vee \neg X_1) \wedge \neg X_2 = 1) \\
&= (0.1 + 0.8 - 0.1 \cdot 0.8) \cdot 0.3 = 0.246
\end{aligned}
$$

- Consider a dependency between X_0 and X_1 such that $P(X_0=1)=0.9$, $P(X_1=1)=0.2$ and $P(X_1=1 \wedge X_0=1)=0.8$. We further have $P(X_2=1)=0.7$. Calculate $P(Y=1)$.

We first note $P(X_0) \cdot P(X_1) = 0.9 \cdot 0.2 = 0.18 \neq P(X_1 \wedge X_0)$, i.e. the events $X_0=1$ and $X_1=1$ are not independent.

$$
\begin{aligned}
P(X_0 \wedge X_1 \vee X_2 = 1) &= P(X_0 = 1 \wedge X_1 = 1) + P(X_2=1) - P(X_0=1 \wedge X_1=1) \cdot P(X_2=1) \\
&= 0.8 + 0.7 - 0.8 \cdot 0.7 = 0.94
\end{aligned}
$$

Notes:

Probability mass functions, densities, distribution functions

- for **discrete RV** $X : \Omega \to E$ with $X \sim P$ we call P **probablity mass function** (pmf) and we have

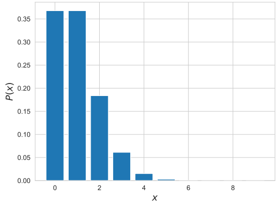
$$\sum_{e \in E} P(X = e) = 1$$

- for **continuous RV** $X : \Omega \to \mathbb{R}$ with $X \sim f$ we call f **probability density function** (pdf) and we have

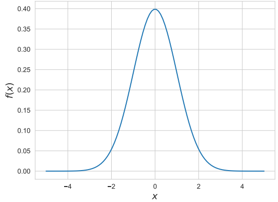
$$\int_{-\infty}^{\infty} f(x) = 1$$

- for pdf f the corresponding **cumulative distribution function** (cdf) is given as

$$F(x) := \int_{-\infty}^{x} f(t)dt$$



discrete **probability mass function**



continuous **probability density function**

Notes:

Mean, variance, and higher moments

- ***n*-th raw moment** $E[X^n]$ is defined as
  - discrete: $E[X^n] := \sum_{k=0}^{\infty} k^n \cdot P(k)$
  - continuous: $E[X^n] := \int_{-\infty}^{\infty} x^n \cdot f(x)dx$
  - first raw moment is **mean** $E[X]$

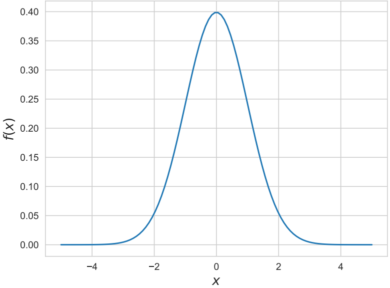
- **variance** of X is defined based on 2nd raw moment

$$\text{Var}(X) := E[(X - E[X])^2] = E[X^2] - E[X]^2$$

- **skewness** of X is defined as

$$\text{Skew}(X) := \frac{1}{\sigma^3} E\left[(X-\mu)^3\right]$$

with $\mu := E(X)$ mean and $\sigma := \sqrt{\text{Var}(X)}$ **standard deviation**



mnemonic for variance

mean of square - square of mean (mossom)

example

for roll of a fair dice, where random variable X maps faces to $\{1, 2, 3, 4, 5, 6\}$ we have
$E(X) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{1}{6} + 6 \cdot \frac{1}{6} = \frac{7}{2}$
$Var(X) = E(X^2) - E(X)^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12}$
$\sigma = \sqrt{Var(X)} \approx 1.71$

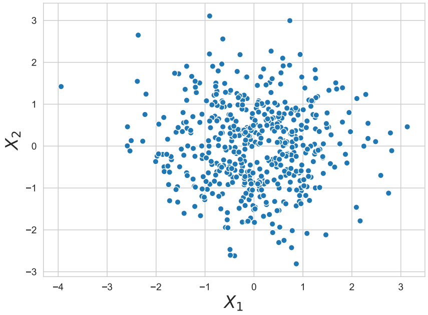
Notes:

Multivariate random variables

- consider vector $\vec{X}$ of **multiple random variables** $X_i : \Omega \to \mathbb{R}$, i.e. $\vec{X} = (X_1, \ldots, X_d)$
- we call $\vec{X} : \Omega^d \to \mathbb{R}^d$ **d-dimensional multivariate random variable**
- $\vec{X} \sim P$ where $P(X_1, \ldots, X_k)$ is **joint distribution** that assigns probabilities to **all combinations of values of X_i**
- $X_i \sim P_i$ where P_i are **marginal distributions** of individual random variables
- X, Y **independent** iff for all x, y we have $P(X = x, Y = y) = P(X = x) \cdot P(Y = y)$
- $\mu = E(\vec{X}) = (E(X_1), \ldots, E(X_2))$ is **mean vector** of multivariate random variable $\vec{X}$

example

500 samples of bivariate random variable $\vec{X} = (X_1, X_2)$ with **Normal marginal distributions** $X_i \sim Norm(0, 1)$
joint distribution is **bivariate Normal distribution** $Norm^2 : \mathbb{R}^2 \to \mathbb{R}$



Notes:

Covariance

covariance of random variables

Let X and Y be two random variables. The covariance $\text{Cov}(X, Y)$ of X and Y is defined as

$$\text{Cov}(X, Y) := E\left[(X - E[X]) \cdot (Y - E[Y])\right]$$

- $\text{Cov}(X, X) = E\left[(X - E[X])^2\right] = \sigma^2 = \text{Var}(X)$
- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$

- $\text{Cov}(X, Y)$ captures **"joint variability" of X and Y**
- **Pearson correlation coefficient** $\rho(X, Y) := \frac{\text{Cov}(X,Y)}{\sqrt{Var(X)}\sqrt{Var(Y)}}$
- X and Y independent $\implies$ $\text{Cov}(X, Y) = 0$ (but not vice-versa) $\rightarrow$ exercise sheet
- for $X_1, \ldots, X_n$ we define symmetric **covariance matrix** as

$$\Sigma(X_1, \ldots, X_n) = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1,X_2) & \ldots & \text{Cov}(X_1,X_n) \\ \vdots & \vdots & \vdots & \vdots \\ \text{Cov}(X_n,X_1) & \text{Cov}(X_n,X_2) & \ldots & \text{Var}(X_n) \end{pmatrix}$$

Notes:

Examples

example 1

500 samples of bivariate random variable (X_1, X_2) with

- $X_1 \sim \text{Norm}(0, 1)$
- $X_2 \sim \text{Norm}(0, 1)$

example 2

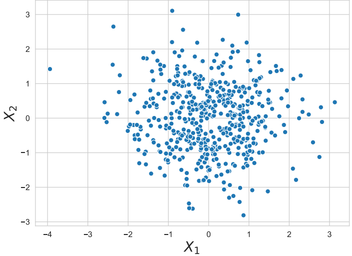
500 samples of bivariate random variable (X_1, X_2) with

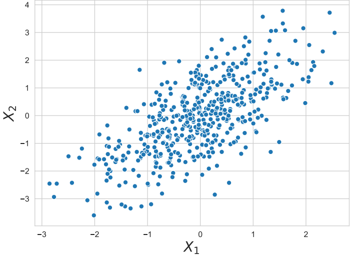
- $X_1 \sim \text{Norm}(0, 1)$
- $X_2 \sim X_1 + \text{Norm}(0, 1)$

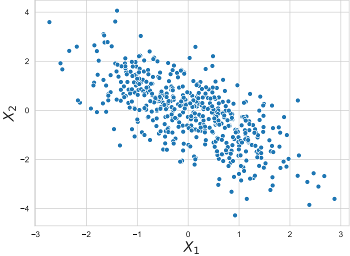
example 3

500 samples of bivariate random variable (X_1, X_2) with

- $X_1 \sim \text{Norm}(0, 1)$
- $X_2 \sim -X_1 + \text{Norm}(0, 1)$







$$\Sigma(X_1, X_2) = \begin{pmatrix} 1.00 & \mathbf{-0.04} \\ \mathbf{-0.04} & 0.99 \end{pmatrix}$$

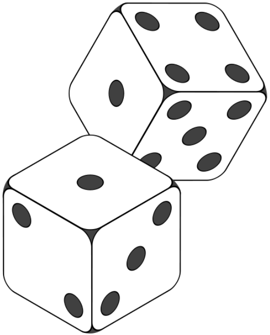
$$\Sigma(X_1, X_2) = \begin{pmatrix} 0.91 & \mathbf{0.88} \\ \mathbf{0.88} & 1.78 \end{pmatrix}$$

$$\Sigma(X_1, X_2) = \begin{pmatrix} 1.02 & \mathbf{-1.02} \\ \mathbf{-1.02} & 2.11 \end{pmatrix}$$

Notes:

In summary

- we introduced basic mathematical concepts to **model uncertainty**
- probabilities can be thought of as representation of our **degree of belief** in an event
- we use uni- and multivariate random variables to **quantitatively model random events**
- basis for **Monte Carlo and Las Vegas algorithms** that utilize randomness and non-determinism
- basis for **statistical modelling and inference** in empirical data



Notes:

Self-study questions

1. List three issues that can introduce uncertainty in real data.
2. What is the difference between an outcome ω and and event E? Is any outcome also an event? Is any event also an outcome?
3. What can you say about the conditional probability $P(B|A)$ if A and B are independent events?
4. What is the difference between independent and mutually exclusive events A and B.
5. Define variance based on the raw moments of a probability distribution.
6. What we can say about two random variables X, Y that exhibit positive covariance.
7. What we can say about two random variables X, Y that exhibit zero covariance.
8. Give an example for two random variables X, Y that exhibit zero covariance but that are not independent.
9. What are the marginal distributions for a multivariate random variable $(X_1, \ldots, X_k)$?
10. How are the mean, variance, and skewness of a random variable related to its raw moments?

Notes:

References

reading list

- G Cardano: **Liber de ludo aleae (Book on Games of Chance)**, 1564
- S Russel, P Norvig: **Artificial Intelligence - A Modern Approach**, 2021
  $\longrightarrow$ Chapter 12
- CM Bishop: **Pattern Recognition and Machine Learning**, Springer, 2006
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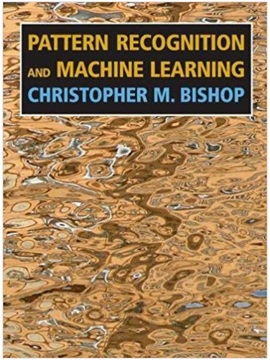

Notes: